

Bash Basics

What is a bash script?

- basic: listing of commands to execute with Bash shell
- Bash is also a programming language - scripts can contain:
 - control structures -- loops, conditionals
 - variables
 - functions
 - arguments (parameters)
 - arrays

Who uses them and why?

- Make life easier: automate or simplify tasks run regularly
- Efficiency - create script to perform repetitive tasks
- Examples:
 - sysadmins needing to check status and running the same commands on a regular basis
 - script to build and deploy personal website

Creating and running

- open, edit, save file with program/list of commands
- convention for bash is to use .sh extension
- change the permissions to make executable
 - `chmod u+x filename`
 - `chmod +x filename`
- `./filename`

Pound-Bang

- aka shebang, hashbang
- first line tells the kernel what program to use to run the script
- Why bother adding to script?
 - add portability - users don't need to know what to use to call script
 - easily run bash scripts from other shells
- Examples with bash:
 - `#!/bin/bash`
 - `#!/user/bin/env bash`

Pound-Bang (continued)

- Can also use with others like Python:
 - `#!/usr/bin/env python3`
- `/usr/bin/env bash` **vs** `/bin/bash`
 - env uses whatever version of the executable comes first in \$PATH
 - env - users can have different behavior

Mac users with homebrew

- use `brew install bash` to get updated version
- different path from pervious slide
 - could be: `/usr/local/bin/bash`

Comments

- # begins a comment from there until end of line
- Exception:
 - pound-bang/shebang on first line of script